

Tucker order-3 vs order-4

node207-23, H100 80GB, BF16 autocast, Llama geometry, seq 1024
16,384 tokens per optimizer step

1 Parameter count

Budget	Layout	Parameters	vs dense
133M	Tucker-4	132,998,816	-124,190,048
133M	Tucker-3: split input	133,002,112	-124,186,752
133M	Tucker-3: split output	132,999,936	-124,188,928
225M	Tucker-4	225,023,872	-32,164,992
225M	Tucker-3: split input	225,000,704	-32,188,160
225M	Tucker-3: split output	224,997,120	-32,191,744

2 Forward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	620.2	50.5
133M	Tucker-3: split input	636.0	103.9
133M	Tucker-3: split output	606.3	93.1
225M	Tucker-4	610.9	59.1
225M	Tucker-3: split input	614.8	112.7
225M	Tucker-3: split output	640.5	104.0

3 Backward, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	796.5	89.9
133M	Tucker-3: split input	743.5	151.9
133M	Tucker-3: split output	728.5	175.0
225M	Tucker-4	778.6	116.3
225M	Tucker-3: split input	701.9	179.7
225M	Tucker-3: split output	771.7	210.6

4 Optimizer, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	83.4	68.3
133M	Tucker-3: split input	85.8	67.5
133M	Tucker-3: split output	81.9	60.5
225M	Tucker-4	92.4	70.1
225M	Tucker-3: split input	83.4	73.8
225M	Tucker-3: split output	97.4	88.9

5 Retraction, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	20.2	21.6
133M	Tucker-3: split input	95.8	95.8
133M	Tucker-3: split output	110.3	109.9
225M	Tucker-4	27.3	28.9
225M	Tucker-3: split input	383.0	384.3
225M	Tucker-3: split output	364.6	363.4

6 Full step, ms

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	1524.9	231.1
133M	Tucker-3: split input	1564.0	419.9
133M	Tucker-3: split output	1527.4	439.8
225M	Tucker-4	1511.4	275.7
225M	Tucker-3: split input	1783.6	752.2
225M	Tucker-3: split output	1879.8	767.8

7 Forward+backward peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	4.19	9.72
225M	Tucker-3: split output	4.20	9.70

8 Full-step peak allocated, GB

Budget	Layout	MB 1×16	MB 16×1
133M	Tucker-4	3.09	8.95
133M	Tucker-3: split input	3.07	8.95
133M	Tucker-3: split output	3.07	8.95
225M	Tucker-4	4.21	9.90
225M	Tucker-3: split input	5.60	9.72
225M	Tucker-3: split output	4.51	9.70

9 Fastest order-3 full step

Budget	MBxAcc	Layout	ms	time/Tucker-4
133M	1x16	Tucker-3: split output	1527.4	1.002x
133M	16x1	Tucker-3: split input	419.9	1.817x
225M	1x16	Tucker-3: split input	1783.6	1.180x
225M	16x1	Tucker-3: split input	752.2	2.728x

10 Run

Field	Value
Commit	0a92e08e9bc62fd1d9485b7bf03d23b905524dc5
Run	node207-23/gpu7
GPU	NVIDIA H100 80GB HBM3
PyTorch	2.9.1+cu128
CUDA	12.8
Samples	3 warmup + 12 measured

Order-3 is represented by three trainable factors and one fixed singleton buffer. The unsplit side uses the generic GEMM fallback in the current prototype. Ratios above 1 mean slower than Tucker-4.